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Properties of Oriens-Lacunosum Moleculare Neurons: Location, Structure, Function, and Molecular Characterization

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Introduction

Oriens-lacunosum moleculare (O-LM) neurons are a specific type of GABAergic interneuron found in the hippocampus that play key roles in regulating neural circuits and memory processing. These neurons have distinct anatomical, physiological, and molecular properties that make them important for controlling hippocampal network activity. (LLM Memory)

Oriens-lacunosum moleculare (O-LM) neurons represent a well-characterized subtype of GABAergic interneurons within the hippocampal formation that have garnered significant attention due to their unique properties and functional roles (Model-Generated). These neurons are distinguished by their specific anatomical location, with cell bodies primarily situated in the stratum oriens of hippocampal areas CA1 and CA3, and axonal projections that extend to the lacunosum-moleculare layer (Model-Generated). O-LM neurons serve as crucial modulators of hippocampal network activity through their inhibitory connections and are particularly important for regulating information flow between different hippocampal layers and controlling the timing of neural activity during memory-related processes (Model-Generated). Their distinctive morphological features, including horizontal dendrites in the oriens layer and characteristic axonal branching patterns, combined with specific electrophysiological properties and molecular marker expression, have made them a valuable model for understanding interneuron diversity and function in the brain (Model-Generated). Recent advances in single-cell transcriptomics and molecular profiling have further refined our understanding of O-LM neuron heterogeneity and their precise roles within hippocampal microcircuits (Model-Generated).

Anatomical Location and Morphology

O-LM neurons have a highly stereotyped morphology with cell bodies located in the stratum oriens of the hippocampus and horizontal dendrites, while their axons project vertically through multiple layers to extensively branch in the stratum lacunosum-moleculare layer. This distinctive anatomical arrangement was first described by Santiago Ramón y Cajal in 1893 and positions these neurons to act as gatekeepers controlling information flow in hippocampal circuits. (13 sources)

O-LM neurons exhibit a remarkably consistent and distinctive morphological organization that has been recognized since the earliest anatomical descriptions of the hippocampus. These neurons were first identified by Santiago Ramón y Cajal in 1893 as large somata neurons located in the hippocampal stratum oriens and sending prominent axonal projections to stratum lacunosum-moleculare (Hilscher et al., 2023).

The cell bodies of O-LM neurons are characteristically located in the deepest layer of the CA1 hippocampal region, specifically in the stratum oriens near the oriens/alveus border, where they display large, horizontally-oriented somata (Nichol et al., 2018) (Hooft et al., 2000) (Hewitt et al., 2021) (Lacaille et al., 1987).

The dendritic architecture of O-LM neurons is highly specialized, with dendrites extending horizontally through the stratum oriens, running perpendicular to the dendrites of pyramidal cells along the CA3-subicular axis (Hewitt et al., 2021) (Zhang et al., 2025). These horizontal dendrites are often densely decorated with long spines and remain confined to the stratum oriens layer (Zemankovics et al., 2010) (Hurtado-Zavala et al., 2017). This dendritic organization allows O-LM neurons to receive primarily local excitatory input from CA1 pyramidal cells and other sources within the oriens layer (Bohm et al., 2015).

The axonal projection pattern of O-LM neurons represents their most defining morphological feature and the basis for their functional role. The axon typically originates from either the soma or a proximal dendrite and projects vertically through the stratum pyramidale and radiatum layers without significant branching (Zemankovics et al., 2010) (Hewitt et al., 2021). Upon reaching the stratum lacunosum-moleculare, the axon undergoes extensive ramification, creating dense terminal arborizations with heavily packed varicosities that form synaptic contacts with the distal dendrites of CA1 pyramidal cells (Zemankovics et al., 2010) (Sekulic et al., 2014) (Sik et al., 1995). Some axon collaterals with boutons may also be observed within the stratum oriens, indicating local inhibitory connections (Zemankovics et al., 2010).

This stereotyped morphological arrangement positions O-LM neurons as critical components of hippocampal feedback inhibitory circuits. Through their anatomical organization, O-LM neurons coordinate a canonical circuit where CA1 pyramidal neurons activate O-LM cells, which in turn provide selective inhibition to the distal dendrites of pyramidal cells in the lacunosum-moleculare layer (Hewitt et al., 2021) (Leao et al., 2012). This strategic positioning allows them to gate incoming sensory information from the entorhinal cortex via the temporoammonic pathway, which terminates precisely in the stratum lacunosum-moleculare where O-LM axons arborize (Zhang et al., 2025) (Muller et al., 2014).

Evidence

(Hilscher et al., 2023)

The alpha2 nicotinic acetylcholine receptor, a subunit with unique and selective expression in inhibitory interneurons associated with principal cells.

"OLM cells were first described by Santiago Ramon y Cajal in 1893 as large somata neurons located in the hippocampal stratum oriens and sending prominent axonal projections to stratum lacunosum moleculare, thus the name OLM."

(Nichol et al., 2018)

Electrophysiological and Morphological Characterization of Chrna2 Cells in the Subiculum and CA1 of the Hippocampus: An Optogenetic Investigation

"The nicotinic acetylcholine receptor alpha2 subunit (Chrna2) is a specific marker for oriens lacunosum-moleculare (OLM) interneurons in the dorsal CA1 region of the hippocampus"

"These CA1 GABAergic, somatostatin (Som)-expressing interneurons are named for their distinctive morphology: their soma and dendritic trees are located in the stratum oriens and their axons extend directly out to arborize in the stratum lacunosum-moleculare (SLM; Cajal, 1911;(McBain et al., 1994)(Sik et al., 1995)(Maccaferri et al., 2000)(Losonczy et al., 2002)(Leão et al., 2012)"

"Chrna2 was also found to be expressed in Som-expressing cells of the subiculum (Leão et al., 2012)"

"We found that subiculum and CA1 dorsal Chrna2 cells similarly expressed the marker somatostatin and had comparable membrane and firing properties. The somas of Chrna2 cells in both regions were found in the deepest layer with axons projecting superficially. However, subiculum Chrna2 cells displayed more extensive projections with dendrites which occupied a significantly larger area than in CA1. The post-synaptic responses elicited by Chrna2 cells in pyramidal cells of both regions revealed comparable inhibitory responses elicited by GABAA receptors and, interestingly, GABAB receptor mediated components."

(Hooft et al., 2000)

Differential Expression of Group I Metabotropic Glutamate Receptors in Functionally Distinct Hippocampal Interneurons

"Type I interneurons are located at the oriens-alveus border and have large, horizontally located cell bodies. Their dendritic trees run horizontally, and their axons run vertically toward stratum lacunosum-moleculare (Fig. 2 A). These interneurons contain somatostatin (Fig. 2 B). Therefore, these interneurons appear identical to the horizontal O-LM cells (Freund et al., 1998)(Katona et al., 1999)"

"Type I interneurons had large horizontally oriented cell somata located at the border of stratum oriens and the alveus, indicating that these cells were most likely identical with the previously described somatostatin-positive oriens-lacunosum moleculare (O-LM) cells (Freund et al., 1998). Reconstruction of type I interneurons revealed their horizontally oriented dendritic tree in stratum oriens and their axonal arborizations in stratum lacunosum-moleculare (n = 5) (Fig. 2 A), and in situ hybridization for somatostatin showed that four of four cells were indeed positive for somatostatin (Fig. 2 B)"

"Type I interneurons responded with a large inward current of −224 pA, were positive for somatostatin, and the majority expressed both mGluR1 and mGluR5"

"Activation of group I mGluRs under current-clamp conditions increased spike frequency and resulted in rhythmic firing activity in type I and II, but not in type III and IV, interneurons."

(Hewitt et al., 2021)

High and low expression of the hyperpolarization activated current (I h) in mouse CA1 stratum oriens interneurons

"OLM cells have a stereotyped morphology with dendrites extending perpendicular to pyramidal cell dendrites along the CA3-subicular axis and an axon that projects through the stratum pyramidale and radiatum until it profusely branches within stratum lacunosum moleculare (Lacaille et al., 1987)(Lacaille et al., 1990)(Maccaferri, 2005)(Maccaferri et al., 1996). Through this anatomical arrangement, OLM cells coordinate a canonical feedback inhibitory circuit where CA1 neurons activate OLM cells which in turn inhibit the distal dendrites of CA1 pyramidal cells to gate incoming information from the entorhinal cortex via the temporoammonic pathway (Blasco-Ibáñez et al., 1995)(Klausberger, 2009)(Leão et al., 2012)(Maccaferri et al., 1995)(Müller et al., 2014). OLM cells exhibit tonic or mildly adapting firings characteristics and display sag and rebound responses during the onset and offset of hyperpolarization due to the expression of h-channels (Lupica et al., 2001)(Maccaferri et al., 1996)."

(Lacaille et al., 1987)

Local circuit interactions between oriens/alveus interneurons and CA1 pyramidal cells in hippocampal slices: electrophysiology and morphology

"Electrophysiological and anatomical techniques were used to determine the role, in the hippocampal circuitry, of local circuit neurons located at the oriens/alveus border (O/A interneurons). Intracellular recording from these cells showed that their response characteristics were clearly nonpyramidal: high input resistance, short membrane time constant, short-duration action potential, pronounced, brief afterhyperpolarizations (AHP), and nondecremental firing during intrasomatic depolarizing current pulses. Intracellular Lucifer yellow (LY) injection and subsequent fluorescence microscopy confirmed their nonpyramidal nature. O/A interneuron somata were bipolar or multipolar; their dendrites projected mostly parallel to the alveus, except for 1 or 2 processes that turned perpendicularly, and ascended through stratum oriens and pyramidale and into radiatum. Their axons were seen to branch profusely in stratum oriens and pyramidale. Simultaneous intracellular recordings from O/A interneurons and CA 1 pyramidal cells showed that pyramidal cells directly excite these interneurons. Major hippocampal afferents also directly excited the O/A interneurons. In a small number of interneuron-pyramidal pairs, stimulation of the O/A interneuron directly inhibited pyramidal cells. In one case, reciprocal connections were observed: The pyramidal cell excited the interneuron, and the interneuron inhibited the pyramidal cell. In 1 interneuron-to- interneuron pair, an inhibitory connection from O/A interneuron to stratum pyramidale interneuron was also observed. With intracellular HRP injections into O/A interneurons and subsequent electron microscopy, we observed that O/A interneuron axons made contacts with pyramidal and nonpyramidal cells. HRP-filled symmetric synaptic contacts were found on pyramidal cell dendrites and somata. HRP-filled axons also made contacts with pyramidal cell initial segments. HRP- filled O/A interneuron axon contacts were also found on nonpyramidal cell dendrites in stratum oriens. These electrophysiological and anatomical results suggest that O/A interneurons make synaptic contact with pyramidal cells and may mediate feedforward and feedback inhibition onto CA 1 pyramidal cells."

(Zhang et al., 2025)

Rethinking parvalbumin: From passive marker to active modulator of hippocampal circuits

"OLM cells have a distinct morphology and circuit position. Although canonically identified by the expression of somatostatin (SST), the OLM cell class is molecularly heterogeneous and contains a functionally distinct subpopulation that co-expresses PV (Fernández et al., 2024)(Que et al., 2021). This distinction is critical, as SST-positive OLM cells are primarily associated with theta rhythms, whereas the PV-positive subset can fire in the gamma range, suggesting a different role in network coordination. OLM cell bodies reside in the outermost layer of the hippocampus (stratum oriens), and they project a long axon up to the innermost layer (stratum lacunosum-moleculare, SLM), the most distal dendritic layer of pyramidal cells (Leão et al., 2012). This is precisely where the main excitatory input from the entorhinal cortex arrives via the perforant path. By selectively inhibiting these distal dendrites, OLM cells act as gatekeepers, controlling the flow of cortical information into the hippocampus and modulating dendritic integration and plasticity (Leão et al., 2012). They are critically involved in the generation of and phase-locking to network theta oscillations

(4-12 Hz) (Amilhon et al., 2015)."

(Zemankovics et al., 2010)

Differences in subthreshold resonance of hippocampal pyramidal cells and interneurons: the role of h-current and passive membrane characteristics

"oriens-lacunosum-moleculare (OLM) cells also had both the cell body and dendritic tree in the stratum oriens, but their horizontally running dendrites were often densely decorated with long spines. Their axon frequently originated from a proximal dendrite, and after ramification the main axon without boutons could be followed into the stratum lacunosum-moleculare. In this layer the axon ramified extensively bearing heavily packed varicosities. Some axon collaterals with boutons were also observed in the stratum oriens."

(Hurtado-Zavala et al., 2017)

TRPV1 regulates excitatory innervation of OLM neurons in the hippocampus

"These neurons have large somata with axons extending to the stratum lacunosum moleculare (Mu□ llet et al., 2014)(Cajal, 0) , and horizontal dendrites projecting through the stratum oriens as well as to further layers of the CA1 region (Freund et al., 1998) . Indeed, we found that most TRPV1-expressing neurons in the stratum oriens also expressed somatostatin"

"TRPV1 is expressed in oriens-lacunosum-moleculare neurons"

"these markers co-localized in a sub-population of presumptive OLM cells in the stratum oriens"

"We found that 81% of TRPV1-expressing neurons in the stratum oriens are densely decorated by detectable mGluR7 signal"

"indicating that these neurons correspond to OLM neurons in the hippocampus. Most TRPV1-positive neurons were also positive for the inhibitory interneuron marker GAD65"

(Bohm et al., 2015)

Serotonin Attenuates Feedback Excitation onto O-LM Interneurons

"O-LM interneurons were identified by their location in stratum oriens, the characteristic elongated shape of their somata and the horizontally oriented dendrites as well as their electrophysiological properties: the typical firing pattern in response to depolarizing current pulses and the prominent sag potential in response to hyperpolarizing current pulses"

"A subset of the recorded cells, identified by the above described criteria, were further validated by biocytin stainings in which we could identify the vertically projecting axons with an extensive arborization in stratum lacunosum-moleculare"

"These interneurons are thought to receive primarily local excitatory input and are, via their axonal projections to stratum lacunosum-moleculare, ideally suited to control entorhinal cortex input"

"GABAergic inhibitory oriens lacunosum-moleculare (O-LM) cells in the hippocampal area CA1 of the rat"

(Sekulic et al., 2014)

Using Multi-Compartment Ensemble Modeling As an Investigative Tool of Spatially Distributed Biophysical Balances: Application to Hippocampal Oriens-Lacunosum/Moleculare (O-LM) Cells

"oriens-lacunosum/moleculare (O-LM) cell located in the CA1 region plays an important role in information flow, neuronal rhythms, and synaptic plasticity (Bartos et al., 2011)(Leão et al., 2012)(Perez et al., 2001)(Matt et al., 2010). O-LM cells possess a distinct morphology. Their somata and dendrites are located in the stratum oriens layer. In contrast, their dense axonal arborizations project to the lacunosum/moleculare layer, synapsing onto distal dendrites of pyramidal cells, where they are thought to influence the efficacy of perforant path input (Sik et al., 1995)[18]. O-LM cells have a variety of voltage-gated ion channels across their somatodendritic and axonal compartments. In particular, one hallmark feature of O-LM cells is that they exhibit a "sag" response to hyperpolarizing current steps, which indicates the presence of hyperpolarization-activated cation currents or I h (Maccaferri et al., 1996). I h contributes to the spontaneous firing of O-LM cells allowing them to play a pacemaker role, and models have suggested how they could contribute to the generation of population theta activities"

(Sik et al., 1995)

Hippocampal CA1 interneurons: an in vivo intracellular labeling study

"Fast spiking interneurons in the CA1 area of the dorsal hippocampus were recorded from and filled with biocytin in anesthetized rats. The full extent of their dendrites and axonal arborizations as well as their calcium binding protein content were examined. Based on the spatial extent of axon collaterals, local circuit cells (basket and O- LM neurons) and long-range cells (bistratified, trilaminar, and backprojection neurons) could be distinguished. Basket cells were immunoreactive for parvalbumin and their axon collaterals were confined to the pyramidal layer. A single basket cell contacted more than 1500 pyramidal neurons and 60 other parvalbumin-positive interneurons. Commissural stimulation directly discharged basket cells, followed by an early and late IPSPs,

indicating interneuronal inhibition of basket cells. The dendrites of another local circuit neuron (O-LM) were confined to stratum oriens and it had a small but high-density axonal terminal field in stratum lacunosum-moleculare. The fastest firing cell of all interneurons was a calbindin-immunoreactive bistratified neuron with axonal targets in stratum oriens and radiatum. Two neurons with their cell bodies in the alveus innervated the CA3 region (backprojection cells), in addition to rich axon collaterals in the CA1 region. The trilaminar interneuron had axon collaterals in strata radiatum, oriens and pyramidale with its dendrites confined to stratum oriens. Commissural stimulation evoked an early EPSP-IPSP-late depolarizing potential sequence in this cell. All interneurons formed symmetric synapses with their targets at the electron microscopic level. These findings indicate that interneurons with distinct axonal targets have differential functions in shaping the physiological patterns of the CA1 network."

(Leao et al., 2012)

OLM interneurons differentially modulate CA3 and entorhinal inputs to hippocampal CA1 neurons

"The vast diversity of GABAergic interneurons is believed to endow hippocampal microcircuits with the required flexibility for memory encoding and retrieval. However, dissection of the functional roles of defined interneuron types has been hampered by the lack of cell-specific tools. We identified a precise molecular marker for a population of hippocampal GABAergic interneurons known as oriens lacunosum-moleculare (OLM) cells. By combining transgenic mice and optogenetic tools, we found that OLM cells are important for gating the information flow in CA1, facilitating the transmission of intrahippocampal information (from CA3) while reducing the influence of extrahippocampal inputs (from the entorhinal cortex). Furthermore, we found that OLM cells were interconnected by gap junctions, received direct cholinergic inputs from subcortical afferents and accounted for the effect of nicotine on synaptic plasticity of the Schaffer collateral pathway. Our results suggest that acetylcholine acting through OLM cells can control the mnemonic processes executed by the hippocampus."

(Muller et al., 2014)

Dendritic inhibition mediated by O-LM and bistratified interneurons in the hippocampus

"In the CA1 region of the hippocampus pyramidal neurons and GABAergic interneurons form local microcircuits. CA1 interneurons are a diverse group consisting of many subtypes, some of which provide compartment-specific inhibition specifically onto pyramidal neuron dendrites. In fact, the majority of inhibitory synapses on pyramidal neurons is found on their dendrites. The specific role of a dendrite-innervating interneuron subtype is primarily determined by its innervation pattern on the distinct dendritic domains of pyramidal neurons. The efficacy of dendritic inhibition in reducing dendritic excitation depends on the relative timing and location of the activated excitatory and inhibitory synapses. In vivo, synaptic properties such as short-term plasticity and neuro-modulation by the basal forebrain, govern the degree of inhibition in distinct dendritic domains in a dynamic, behavior dependent manner, specifically during network oscillation such as the theta rhythm. In this review we focus on two subtypes of dendrite-innervating interneurons: the oriens-lacunosum moleculare (O-LM) interneuron and the bistratified interneuron. Their molecular marker profile, morphology, and function in vivo and in vitro are well studied. We strive to integrate this diverse information from the cellular to the network level, and to provide insight into how the different characteristics of O-LM and bistratified interneurons affect dendritic excitability, network activity, and behavior."

Projection Patterns and Connectivity

O-LM neurons exhibit complex connectivity patterns that position them as key regulators of hippocampal information flow, receiving excitatory inputs from local CA1 pyramidal cells and diverse sources while providing selective inhibition to pyramidal cell distal dendrites. They also receive cholinergic inputs from the medial septum and are interconnected through gap junctions, allowing them to coordinate network activity and gate sensory information from the entorhinal cortex. (8 sources)

The connectivity patterns of O-LM neurons reflect their specialized role as circuit modulators within hippocampal networks. These neurons receive diverse synaptic inputs that enable them to integrate information from multiple sources and coordinate appropriate inhibitory responses. O-LM cells receive inputs from local CA1 pyramidal cells, bistratified cells, interneuron-specific cells, and long-range projecting GABAergic inputs from the medial septum (Guet-McCreight et al., 2020). This diverse input pattern allows O-LM neurons to respond to both local circuit activity and broader network states influenced by subcortical modulation.

A particularly important aspect of O-LM neuron connectivity involves their cholinergic innervation from the medial septum. O-LM interneurons display fast nicotinic excitation in response to cholinergic input from the medial septum, and this excitation is sufficient to drive these neurons above action potential threshold (Muller et al., 2017) (Leao et al., 2012). The medial septum and diagonal band of Broca provide cholinergic, GABAergic, and glutamatergic inputs across the hippocampal axis, likely influencing disinhibition and oscillatory activity during various behavioral states (Thulin et al., 2025). Cholinergic excitation of

somatostatin-positive putative O-LM interneurons occurs during the association of multisensory contextual input with an aversive stimulus (Muller et al., 2017).

O-LM neurons are also interconnected through gap junctions, which allows for coordinated activity across the population (Muller et al., 2017) (Leao et al., 2012). This electrical coupling enables synchronized responses and may contribute to their role in generating and maintaining network oscillations.

The output connectivity of O-LM neurons is highly specific and functionally significant. These neurons innervate only the apical tuft of pyramidal cells in stratum lacunosum-moleculare while receiving inputs only in stratum oriens (Tecuatl et al., 2020) (Zemankovics et al., 2010) (McBain et al., 1994). Through their axonal projections to the stratum lacunosum-moleculare, O-LM cells synapse onto the distal dendrites of pyramidal cells, which allows them to gate the flow of sensory information from entorhinal cortex (Guet-McCreight et al., 2020).

Recent evidence reveals that O-LM cell connectivity varies along the dorsal-ventral axis of the hippocampus. CA1 and CA3 inputs differ between hippocampal poles, suggesting that O-LM cells may have distinct feedback and feed-forward inhibitory roles in different parts of the hippocampal microcircuit. Intermediate O-LM cells uniquely receive substantial input from the subiculum and dorsal/medial raphe nuclei, as well as widespread CA2 inputs potentially linked to social memory (Thulin et al., 2025). Additionally, excitatory input to intermediate-ventral O-LM cells partly arises from CA1 projection neurons targeting the medial prefrontal cortex, suggesting these neurons may be involved in hippocampal-cortical communication (Thulin et al., 2025).

The dense axonal output of O-LM neurons shows remarkable layer-specificity, with a sharp restriction of their dense axonal plexus to the lacunosum-moleculare layer without extension into the immediately adjacent dentate molecular layer (Peng et al., 2013) (McBain et al., 1994). This precise laminar targeting is characteristic of many classes of GABAergic interneurons in the hippocampal formation and reflects the specialized functional roles of different interneuron types (Peng et al., 2013).

Evidence

(Guet-McCreight et al., 2020)

Computationally going where experiments cannot: a dynamical assessment of dendritic ion channel currents during in vivo-like states

> *"In CA1, OLM cells have somata and dendrites in the stratum oriens/alveus, and receive inputs from local pyramidal cells 32 , bistratified cells 32 , interneuron specific 3 cells 33 , and long-range projecting GABAergic inputs from medial septum 33 , among others. They have axons projecting to the stratum lacunosum moleculare where they synapse onto the distal dendrites of pyramidal cells, which allows them to gate the flow of sensory information from entorhinal cortex 32,34"*

(Muller et al., 2017)

Septo–hippocampal interaction

> *"oriens-lacunosum moleculare (O-LM) interneurons, with their somata and dendrites located in stratum oriens and their axonal projections in stratum radiatum and lacunosum moleculare, display fast nicotinergic excitation in response to cholinergic input from the medial septum (Leão et al., 2012). The excitation of this interneuron sub-type is thought to result in a strong inhibition of distal pyramidal neuron dendritic tufts located in stratum lacunosum moleculare, which counteracts the excitation from the entorhinal cortex conveyed via the temporoammonic pathway (Fuhrmann et al., 2015)(Leão et al., 2012)"*

> *"cholinergic septal inputs indeed excite the hippocampal O-LM interneurons sufficiently to cross the action potential threshold (Lovett-Barron et al., 2014). Cholinergic excitation of somatostatin-positive putative O-LM interneurons occurs during the association of multisensory contextual input with an aversive stimulus (Lovett-Barron et al., 2014)."*

(Leao et al., 2012)

OLM interneurons differentially modulate CA3 and entorhinal inputs to hippocampal CA1 neurons

> *"The vast diversity of GABAergic interneurons is believed to endow hippocampal microcircuits with the required flexibility for memory encoding and retrieval. However, dissection of the functional roles of defined interneuron types has been hampered by the*

lack of cell-specific tools. We identified a precise molecular marker for a population of hippocampal GABAergic interneurons known as oriens lacunosum-moleculare (OLM) cells. By combining transgenic mice and optogenetic tools, we found that OLM cells are important for gating the information flow in CA1, facilitating the transmission of intrahippocampal information (from CA3) while reducing the influence of extrahippocampal inputs (from the entorhinal cortex). Furthermore, we found that OLM cells were interconnected by gap junctions, received direct cholinergic inputs from subcortical afferents and accounted for the effect of nicotine on synaptic plasticity of the Schaffer collateral pathway. Our results suggest that acetylcholine acting through OLM cells can control the mnemonic processes executed by the hippocampus."

(Thulin et al., 2025)

Differentiated Presynaptic Input to OLM-2 Cells Along the Hippocampal Dorsoventral Axis: Implications for Hippocampal Microcircuit Function

"Oriens lacunosum-moleculare (OLM) interneurons, identified by the Chrna2 gene"

"The Chrna2 gene expression is restricted to the stratum oriens in the hippocampus in both rats and mice (Ishii et al., 2005) and is specifically expressed in a subset of CA1 hippocampal interneurons, the oriens lacunosummoleculare (OLM) cells (Leão et al., 2012). Traditionally, OLM cells have been identified through their expression of somatostatin (Sst). However, in-depth single-cell transcriptomic cluster analysis has unveiled at least 11 distinct subpopulations of Sst-expressing interneurons (2017). Within these clusters, various classes of interneurons were identified, including back projecting, hippocampo-septal, oriens-bistratified, and OLM cells. Among these clusters, OLM cells were classified into a Sst and Prepronociceptin (Pnoc) co-expressing group (further divided into three subclusters)"

"OLM-2 cells receive differential innervation across the dorsal, intermediate, and ventral hippocampus"

"CA1 and CA3 inputs differ between hippocampal poles, suggesting that OLM-2 cells may have distinct feedback and feed-forward inhibitory roles in the hippocampal microcircuit. Intermediate OLM-2 cells uniquely receive substantial input from the subiculum and dorsal/medial raphe nuclei, as well as widespread CA2 inputs potentially linked to social memory. The medial septum and diagonal band of Broca provide cholinergic, GABAergic, and glutamatergic inputs across the axis, likely influencing disinhibition and oscillatory activity during various behavioral states. Excitatory input to intermediate-ventral OLM-2 cells partly arises from CA1 projection neurons targeting the mPFC."

(Tecuatl et al., 2020)

Comprehensive Estimates of Potential Synaptic Connections in Local Circuits of the Rodent Hippocampal Formation by Axonal-Dendritic Overlap

"CA1 oriens-lacunosum moleculare (O-LM) interneurons innervate only the apical tuft of pyramidal cells (PCs) in stratum lacunosum-moleculare (SLM) and receive inputs only in stratum oriens (SO) (McBain et al., 1994)(Losonczy et al., 2002)(Zemankovics et al., 2010)."

(Zemankovics et al., 2010)

Differences in subthreshold resonance of hippocampal pyramidal cells and interneurons: the role of h-current and passive membrane characteristics

"oriens-lacunosum-moleculare (OLM) cells also had both the cell body and dendritic tree in the stratum oriens, but their horizontally running dendrites were often densely decorated with long spines. Their axon frequently originated from a proximal dendrite, and after ramification the main axon without boutons could be followed into the stratum lacunosum-moleculare. In this layer the axon ramified extensively bearing heavily packed varicosities. Some axon collaterals with boutons were also observed in the stratum oriens."

(McBain et al., 1994)

Activation of metabotropic glutamate receptors differentially affects two classes of hippocampal interneurons and potentiates excitatory synaptic transmission

"Based on responses to metabotropic glutamate receptor (mGluR) activation, we have characterized two distinct classes of interneuron in stratum (st.) oriens of the CA1 region of hippocampus. One type of interneuron was strongly excited by 1S,3R-aminocyclopentane dicarboxylic acid (ACPD), responding with a large inward current accompanied by increased baseline noise and prominent current oscillations. A second interneuron population responded with a modest inward current with no changes in baseline noise. These two classes of responses persisted in the presence of tetrodotoxin and antagonists of ionotropic glutamate and GABA receptors, suggesting that the inward currents result from mGluRs on the interneurons themselves. The two physiologically defined cell types correspond to two distinct morphological cell types in st. oriens/alveus, distinguished by very different patterns of local axonal connections. Large oscillatory inward current responses were recorded predominantly from an interneuron type whose axons heavily innervated st. lacunosum. The more modest inward current response was generally found in interneurons whose axons innervated the somata and proximal dendrites of CA1 pyramidal neurons. These differences in physiology and local circuitry imply that activation of mGluRs in st. oriens will cause very strong excitation of interneurons synapsing in st. lacunosum, and weaker excitation of interneurons innervating pyramidal cells at the soma and proximal dendrites.

These data suggest that each interneuron population has a specific role in hippocampal function, and that mGluR activation will affect the local circuit differently for each interneuron type. Metabotropic GluR activation also markedly enhanced the amplitudes of the evoked and spontaneous EPSCs received by all interneurons in the region, independent of changes in the postsynaptic holding current and with no change in the kinetics of the EPSC. In contrast to the enhancement of evoked and spontaneous EPSCs, miniature EPSCs recorded in the presence of tetrodotoxin were not increased. These data suggest that ACPD acts at a presynaptic site to potentiate the EPSC. Taken together, these results highlight an important modulatory role for metabotropic receptors located at sites both pre- and postsynaptic to CA1 st. oriens interneurons."

(Peng et al., 2013)

A Reorganized GABAergic Circuit in a Model of Epilepsy: Evidence from Optogenetic Labeling and Stimulation of Somatostatin Interneurons

"Classical oriens lacunosum-moleculare (O-LM) cells (Lorente de Nó , 1934;(McBain et al., 1994)Sik et al., 1995) are likely to be the major source of the reorganized fibers, based on the exuberant ChR2-eYFP-labeled axonal plexus in s. lacunosum-moleculare in normal and control mice"

"A remarkable feature of O-LM and other SOM-expressing neurons that project to s. lacunosum-moleculare of CA1 is the sharp restriction of their dense axonal plexus to this layer, without extension into the immediately adjacent dentate molecular layer (Klausberger, 2009)"

"the laminar-specific distribution of these axons, characteristic of many classes of GABAergic interneurons in the hippocampal formation (Halasy and (Houser et al., 1990)Klausberger and Somogyi, 2008)"

Electrophysiological Properties and Function

O-LM neurons exhibit distinctive electrophysiological properties including regular-to-fast spiking patterns, pronounced sag responses due to h-channels, and theta frequency resonance that enables them to function as gatekeepers controlling hippocampal information flow and contributing to network oscillations and synaptic plasticity. (16 sources)

O-LM neurons display a characteristic set of electrophysiological properties that distinguish them from other hippocampal interneuron subtypes and enable their specialized functional roles. These neurons exhibit regular-to-fast action potential spiking patterns and show tonic or mildly adapting firing characteristics during sustained depolarization (Friend et al., 2019) (Oren et al., 2009) (Nicholson et al., 2014) (Hewitt et al., 2021) (Lacaille et al., 1987). Some studies have identified subpopulations of O-LM-like neurons that are fast-spiking and show virtually no spike frequency accommodation (Oliva et al., 2000).

A hallmark electrophysiological feature of O-LM neurons is their pronounced sag response to hyperpolarizing current steps, which reflects the presence of hyperpolarization-activated cation currents (Ih) mediated by h-channels (Hewitt et al., 2021) (Sekulic et al., 2020) (Sekulic et al., 2014). These neurons exhibit theta spiking resonance frequencies that depend on their dendrites having hyperpolarization-activated cation channels, with immunohistochemical studies confirming dendritic expression of the HCN2 subunit (Sekulic et al., 2019). The Ih current contributes to the spontaneous firing of O-LM cells, allowing them to play a pacemaker role and potentially contribute to the generation of population theta activities (Sekulic et al., 2014).

O-LM neurons also display distinctive passive membrane properties, including slow membrane time constants and relatively high input resistances (Sekulic et al., 2020). Action potential half-widths are larger and membrane time constants are slower compared to other interneuron subtypes such as parvalbumin-positive interneurons (Sekulic et al., 2020). Earlier characterizations reported high resting membrane potentials, high input resistance, and large amplitude action potentials followed by pronounced afterhyperpolarizations (Lacaille et al., 1988).

Functionally, O-LM neurons serve as critical gatekeepers that control information flow within hippocampal circuits. They operate by gating incoming sensory and ongoing contextual information in the CA1 region, facilitating the transmission of intrahippocampal information from CA3 while reducing the influence of extrahippocampal inputs from the entorhinal cortex (Sekulic et al., 2020) (Leao et al., 2012). Through their selective inhibition of pyramidal cell distal dendrites, O-LM neurons can suppress excitatory input from the entorhinal cortex via the temporoammonic pathway (Hewitt et al., 2021) (Leao et al., 2012) (Muller et al., 2014).

O-LM neurons are critically involved in hippocampal network oscillations, particularly theta rhythms. They fire phase-locked to prominent hippocampal theta rhythms and provide theta-frequency patterned output to distal dendritic segments of pyramidal cells (Sekulic et al., 2020) (Kitchigina, 2010). The neurons fire intermittently at theta frequency and are involved in the generation of and phase-locking to network theta oscillations (Kitchigina, 2010) (Zhang et al., 2025). Interestingly, while somatostatin-positive O-LM cells are primarily associated with theta rhythms, a functionally distinct subpopulation that co-expresses parvalbumin can fire in the gamma range, suggesting different roles in network coordination (Zhang et al., 2025) (Que et al., 2021) (Fernandez et al., 2024).

O-LM neurons exhibit notable synaptic plasticity properties, including a form of long-term potentiation that is independent of NMDA receptors but dependent on Ca^{2+}-permeable AMPA receptors and group I metabotropic glutamate receptors (Friend et al., 2019) (Oren et al., 2009) (Nicholson et al., 2014). This "anti-Hebbian" form of LTP can be induced by presynaptic glutamate release when the postsynaptic membrane potential is hyperpolarized and involves retrograde signaling mechanisms (Oren et al., 2009) (Nicholson et al., 2014). These neurons express both endocannabinoid biosynthetic enzymes and the type I metabotropic glutamate receptors necessary for endocannabinoid production (Friend et al., 2019).

Evidence

(Friend et al., 2019)

Hippocampal Stratum Oriens Somatostatin-Positive Cells Undergo CB1-Dependent Long-Term Potentiation and Express Endocannabinoid Biosynthetic Enzymes

> *"Hippocampal CA1 stratum oriens interneuron subtypes include oriens lacunosum-moleculare (O-LM) interneurons, which can be identified by the expression of somatostatin and have regular-to-fast action potential spiking patterns (Oren et al., 2009)(Nicholson et al., 2014)(Huh et al., 2016). O-LM cell soma and dendrites reside in the stratum oriens and their axons project to the stratum lacunosum-moleculare layer"*

> *"SOM+ neurons have been previously shown to undergo LTP [32] in a manner that is dependent on type I mGluRs [32](Lapointe et al., 2004)(Vasseur et al., 2008)"*

> *"RT-PCR results demonstrate that stratum oriens interneurons express mRNA for both endocannabinoid biosynthetic enzymes and the type I metabotropic glutamate receptors (mGluRs), necessary for endocannabinoid production. Immunohistochemical staining further confirmed the presence of diacylglycerol lipase alpha, an endocannabinoid-synthesizing enzyme, in oriens interneurons"*

> *"Using single-cell RT-PCR, we classified four main groups: SOM+, PV+, calretinin, and calbindin, as previously reported (Freund et al., 1998)(Tamamaki et al., 2003)(Katsumaru et al., 2004)."*

(Oren et al., 2009)

Role of Ionotropic Glutamate Receptors in Long-Term Potentiation in Rat Hippocampal CA1 Oriens-Lacunosum Moleculare Interneurons

> *"Some interneurons of the hippocampus exhibit NMDA receptor-independent long-term potentiation (LTP) that is induced by presynaptic glutamate release when the postsynaptic membrane potential is hyperpolarized. This "anti-Hebbian" form of LTP is prevented by postsynaptic depolarization or by blocking AMPA and kainate receptors. Although both AMPA and kainate receptors are expressed in hippocampal interneurons, their relative roles in anti-Hebbian LTP are not known. Because interneuron diversity potentially conceals simple rules underlying different forms of plasticity, we focus on glutamatergic synapses onto a subset of interneurons with dendrites in stratum oriens and a main ascending axon that projects to stratum lacunosum moleculare, the oriens-lacunosum moleculare (O-LM) cells. We show that anti-Hebbian LTP in O-LM interneurons has consistent induction and expression properties, and is prevented by selective inhibition of AMPA receptors. The majority of the ionotropic glutamatergic synaptic current in these cells is mediated by inwardly rectifying Ca2+-permeable AMPA receptors. Although GluR5-containing kainate receptors contribute to synaptic currents at high stimulus frequency, they are not required for LTP induction. Glutamatergic synapses on O-LM cells thus behave in a homogeneous manner and exhibit LTP dependent on Ca2+-permeable AMPA receptors."*

(Nicholson et al., 2014)

Long-term potentiation in hippocampal oriens interneurons: postsynaptic induction, presynaptic expression and evaluation of candidate retrograde factors

> *"Several types of hippocampal interneurons exhibit a form of long-term potentiation (LTP) that depends on Ca2+-permeable AMPA receptors and group I metabotropic glutamate receptors. Several sources of evidence point to a presynaptic locus of LTP*

maintenance. The retrograde factor that triggers the expression of LTP remains unidentified. Here, we show that trains of action potentials in putative oriens-lacunosum-moleculare interneurons of the mouse CA1 region can induce long-lasting potentiation of stimulus-evoked excitatory postsynaptic currents that mimics LTP elicited by high-frequency afferent stimulation. We further report that blockers of nitric oxide production or TRPV1 receptors failed to prevent LTP induction. The present results add to the evidence that retrograde signalling underlies N-methyl-d-aspartate (NMDA) receptor-independent LTP in oriens interneurons, mediated by an unidentified factor."

(Hewitt et al., 2021)

High and low expression of the hyperpolarization activated current (I h) in mouse CA1 stratum oriens interneurons

"OLM cells have a stereotyped morphology with dendrites extending perpendicular to pyramidal cell dendrites along the CA3-subicular axis and an axon that projects through the stratum pyramidale and radiatum until it profusely branches within stratum lacunosum moleculare (Lacaille et al., 1987)(Lacaille et al., 1990)(Maccaferri, 2005)(Maccaferri et al., 1996). Through this anatomical arrangement, OLM cells coordinate a canonical feedback inhibitory circuit where CA1 neurons activate OLM cells which in turn inhibit the distal dendrites of CA1 pyramidal cells to gate incoming information from the entorhinal cortex via the temporoammonic pathway (Blasco-Ibáńez et al., 1995)(Klausberger, 2009)(Leão et al., 2012)(Maccaferri et al., 1995)(Mu llen et al., 2014). OLM cells exhibit tonic or mildly adapting firings characteristics and display sag and rebound responses during the onset and offset of hyperpolarization due to the expression of h-channels (Lupica et al., 2001)(Maccaferri et al., 1996)."

(Lacaille et al., 1987)

Local circuit interactions between oriens/alveus interneurons and CA1 pyramidal cells in hippocampal slices: electrophysiology and morphology

"Electrophysiological and anatomical techniques were used to determine the role, in the hippocampal circuitry, of local circuit neurons located at the oriens/alveus border (O/A interneurons). Intracellular recording from these cells showed that their response characteristics were clearly nonpyramidal: high input resistance, short membrane time constant, short-duration action potential, pronounced, brief afterhyperpolarizations (AHP), and nondecremental firing during intrasomatic depolarizing current pulses. Intracellular Lucifer yellow (LY) injection and subsequent fluorescence microscopy confirmed their nonpyramidal nature. O/A interneuron somata were bipolar or multipolar; their dendrites projected mostly parallel to the alveus, except for 1 or 2 processes that turned perpendicularly, and ascended through stratum oriens and pyramidale and into radiatum. Their axons were seen to branch profusely in stratum oriens and pyramidale. Simultaneous intracellular recordings from O/A interneurons and CA 1 pyramidal cells showed that pyramidal cells directly excite these interneurons. Major hippocampal afferents also directly excited the O/A interneurons. In a small number of interneuron-pyramidal pairs, stimulation of the O/A interneuron directly inhibited pyramidal cells. In one case, reciprocal connections were observed: The pyramidal cell excited the interneuron, and the interneuron inhibited the pyramidal cell. In 1 interneuron-to- interneuron pair, an inhibitory connection from O/A interneuron to stratum pyramidale interneuron was also observed. With intracellular HRP injections into O/A interneurons and subsequent electron microscopy, we observed that O/A interneuron axons made contacts with pyramidal and nonpyramidal cells. HRP-filled symmetric synaptic contacts were found on pyramidal cell dendrites and somata. HRP-filled axons also made contacts with pyramidal cell initial segments. HRP- filled O/A interneuron axon contacts were also found on nonpyramidal cell dendrites in stratum oriens. These electrophysiological and anatomical results suggest that O/A interneurons make synaptic contact with pyramidal cells and may mediate feedforward and feedback inhibition onto CA 1 pyramidal cells."

(Oliva et al., 2000)

Novel Hippocampal Interneuronal Subtypes Identified Using Transgenic Mice That Express Green Fluorescent Protein in GABAergic Interneurons

"In stratum oriens of the hippocampus, EGFP-expressing interneurons were comprised almost exclusively of oriens/alveus interneurons with lacunosum-moleculare axon arborization (O-LM cells)"

"EGFP was found to be expressed in a subpopulation of somatostatin-containing GABAergic interneurons in the hippocampus and neocortex"

"These newly identified interneuronal subtypes appeared to be closely related to O-LM cell, as they selectively innervate stratum lacunosum-moleculare. Whole-cell patch-clamp recordings revealed that these cells were fast-spiking and showed virtually no spike frequency accommodation. The microanatomical features of these cells suggest that they function primarily as "input-biasing" neurons, in that synaptic volleys in stratum radiatum would lead to their activation, which in turn would result in selective suppression of excitatory input from the entorhinal cortex onto CA1 pyramidal cells."

(Sekulic et al., 2020)

Integration of Within-Cell Experimental Data With Multi-Compartmental Modeling Predicts H-Channel Densities and Distributions in Hippocampal OLM Cells

"The oriens-lacunosum/moleculare (OLM) interneuron operates as a "gate" that controls incoming sensory and ongoing contextual information in the CA1 of the hippocampus"

"OLM cells fire phase-locked to the prominent hippocampal theta rhythms"

"having horizontal cell body and dendrites confined to the oriens layer, and perpendicularly projecting axons which ramify in the lacunosum/moleculare layer"

"SOM-YFP stratum oriens interneurons mice had slow membrane time constants and relatively high input resistances"

"Action potential half-widths were larger and membrane time constants were slower than previously reported for YFP neurons from PV-CRE/Rosa26YFP mice"

"this population had considerable hyperpolarization-induced sag, which, when combined with their higher input resistance, is considered a hallmark feature of OLM cells (Maccaferri and Lacaille, 2003)."

(Sekulic et al., 2014)

Using Multi-Compartment Ensemble Modeling As an Investigative Tool of Spatially Distributed Biophysical Balances: Application to Hippocampal Oriens-Lacunosum/Moleculare (O-LM) Cells

"oriens-lacunosum/moleculare (O-LM) cell located in the CA1 region plays an important role in information flow, neuronal rhythms, and synaptic plasticity (Bartos et al., 2011)(Leão et al., 2012)(Perez et al., 2001)(Matt et al., 2010). O-LM cells possess a distinct morphology. Their somata and dendrites are located in the stratum oriens layer. In contrast, their dense axonal arborizations project to the lacunosum/moleculare layer, synapsing onto distal dendrites of pyramidal cells, where they are thought to influence the efficacy of perforant path input (Sik et al., 1995)[18]. O-LM cells have a variety of voltage-gated ion channels across their somatodendritic and axonal compartments. In particular, one hallmark feature of O-LM cells is that they exhibit a "sag" response to hyperpolarizing current steps, which indicates the presence of hyperpolarization-activated cation currents or I h (Maccaferri et al., 1996). I h contributes to the spontaneous firing of O-LM cells allowing them to play a pacemaker role, and models have suggested how they could contribute to the generation of population theta activities"

(Sekulic et al., 2019)

Somatodendritic HCN channels in hippocampal OLM cells revealed by a convergence of computational models and experiments

"The oriens-lacunosum/moleculare (OLM) interneuron has been implicated as a critical controller of hippocampal memory"

"We performed a set of whole-cell recordings of OLM cells from mouse hippocampus"

"OLM cells exhibit theta spiking resonance frequencies that depend on their dendrites having hyperpolarization-activated cation channels (h-channels)"

"Immunohistochemical localization of the HCN2 subunit confirmed dendritic expression."

(Lacaille et al., 1988)

Stratum lacunosum-moleculare interneurons of hippocampal CA1 region. I. Intracellular response characteristics, synaptic responses, and morphology

"L-M interneurons had a high resting membrane potential (-58 mV), a high input resistance (64 M omega), and a large amplitude (60 mV), relatively long duration (2 msec) action potential. A large afterhyperpolarization (11 mV, 34 msec) followed a single action potential. Most L-M interneurons did not display any spontaneous firing"

"Cells were generally fusiform or multipolar, with aspinous, beaded dendritic processes ramifying in stratum lacunosum-moleculare, radiatum, and (sometimes) oriens. The varicose axon originated from a primary dendrite, projected along stratum lacunosum-moleculare, branched profusely in stratum radiatum, and coursed toward and into stratum pyramidale and occasionally into oriens. Processes of cells with somata in the L-M region of CA1 were not restricted to the CA1 region. The dendritic and axonal processes of some L-M interneurons were seen ascending in stratum lacunosum- moleculare, crossing the hippocampal fissure, and coursing in stratum moleculare of the dentate gyrus"

"The somata were situated in stratum lacunosum-moleculare, near the stratum radiatum border. In general, the cells were fusiform and multipolar, approximately 20 pm in diameter"

"The dendrites were mostly smooth and beaded"

"These dendrites radiated out from the soma along stratum lacunosummoleculare; after a short distance, some turned into stratum radiatum, where they branched"

"In 4 cells, some dendrites crossed the pyramidal cell layer and entered stratum oriens"

"In some interneurons the dendritic processes ran for great distances and expanded outside the CA1 region. For 3 such cells some dendritic branches could be followed ascending in stratum lacunosum-moleculare, crossing the hippocampal fissure and projecting into the dentate gyrus molecular layer"

(Leao et al., 2012)

OLM interneurons differentially modulate CA3 and entorhinal inputs to hippocampal CA1 neurons

> *"The vast diversity of GABAergic interneurons is believed to endow hippocampal microcircuits with the required flexibility for memory encoding and retrieval. However, dissection of the functional roles of defined interneuron types has been hampered by the lack of cell-specific tools. We identified a precise molecular marker for a population of hippocampal GABAergic interneurons known as oriens lacunosum-moleculare (OLM) cells. By combining transgenic mice and optogenetic tools, we found that OLM cells are important for gating the information flow in CA1, facilitating the transmission of intrahippocampal information (from CA3) while reducing the influence of extrahippocampal inputs (from the entorhinal cortex). Furthermore, we found that OLM cells were interconnected by gap junctions, received direct cholinergic inputs from subcortical afferents and accounted for the effect of nicotine on synaptic plasticity of the Schaffer collateral pathway. Our results suggest that acetylcholine acting through OLM cells can control the mnemonic processes executed by the hippocampus."*

(Muller et al., 2014)

Dendritic inhibition mediated by O-LM and bistratified interneurons in the hippocampus

> *"In the CA1 region of the hippocampus pyramidal neurons and GABAergic interneurons form local microcircuits. CA1 interneurons are a diverse group consisting of many subtypes, some of which provide compartment-specific inhibition specifically onto pyramidal neuron dendrites. In fact, the majority of inhibitory synapses on pyramidal neurons is found on their dendrites. The specific role of a dendrite-innervating interneuron subtype is primarily determined by its innervation pattern on the distinct dendritic domains of pyramidal neurons. The efficacy of dendritic inhibition in reducing dendritic excitation depends on the relative timing and location of the activated excitatory and inhibitory synapses. In vivo, synaptic properties such as short-term plasticity and neuro-modulation by the basal forebrain, govern the degree of inhibition in distinct dendritic domains in a dynamic, behavior dependent manner, specifically during network oscillation such as the theta rhythm. In this review we focus on two subtypes of dendrite-innervating interneurons: the oriens-lacunosum moleculare (O-LM) interneuron and the bistratified interneuron. Their molecular marker profile, morphology, and function in vivo and in vitro are well studied. We strive to integrate this diverse information from the cellular to the network level, and to provide insight into how the different characteristics of O-LM and bistratified interneurons affect dendritic excitability, network activity, and behavior."*

(Kitchigina, 2010)

Theta Oscillations and Reactivity of Hippocampal Stratum Oriens Neurons

> *"O-LM interneurons are of great interest since they evidently provide a theta-frequency patterned output to distal dendritic segments of pyramidal cells (Gloveli et al., 2005) that receive specific sensory information (reviewed in Vinogradova (Vinogradova, 1995)) essential for the activity of place cells (Losonczy et al., 2002)"*

> *"oriens-lacunosum moleculare (O-LM) interneurons because they are known to fire intermittently at the theta frequency (Buzsáki, 2002)(Klausberger et al., 2003)"*

> *"GABAergic neurons located in the stratum oriens of the CA1 and CA3 hippocampal regions (Gulyás et al., 2003)(Tóth et al., 1992)"*

(Zhang et al., 2025)

Rethinking parvalbumin: From passive marker to active modulator of hippocampal circuits

> *"OLM cells have a distinct morphology and circuit position. Although canonically identified by the expression of somatostatin (SST), the OLM cell class is molecularly heterogeneous and contains a functionally distinct subpopulation that co-expresses PV (Fernández et al., 2024)(Que et al., 2021). This distinction is critical, as SST-positive OLM cells are primarily associated with theta rhythms, whereas the PV-positive subset can fire in the gamma range, suggesting a different role in network coordination. OLM cell bodies reside in the outermost layer of the hippocampus (stratum oriens), and they project a long axon up to the innermost layer (stratum lacunosum-moleculare, SLM), the most distal dendritic layer of pyramidal cells (Leão et al., 2012). This is precisely where the main excitatory input from the entorhinal cortex arrives via the perforant path. By selectively inhibiting these distal dendrites, OLM cells act as gatekeepers, controlling the flow of cortical information into the hippocampus and modulating dendritic integration and plasticity (Leão et al., 2012). They are critically involved in the generation of and phase-locking to network theta oscillations (4-12 Hz) (Amilhon et al., 2015)."*

(Que et al., 2021)

Transcriptional and morphological profiling of parvalbumin interneuron subpopulations in the mouse hippocampus

> *"The diversity reflected by >100 different neural cell types fundamentally contributes to brain function and a central idea is that neuronal identity can be inferred from genetic information. Recent large-scale transcriptomic assays seem to confirm this hypothesis, but a lack of morphological information has limited the identification of several known cell types. In this study, we used single-cell RNA-seq in morphologically identified parvalbumin interneurons (PV-INs), and studied their transcriptomic states in the morphological, physiological, and developmental domains. Overall, we find high transcriptomic similarity among PV-INs, with few genes showing divergent expression between morphologically different types. Furthermore, PV-INs show a uniform synaptic cell*

adhesion molecule (CAM) profile, suggesting that CAM expression in mature PV cells does not reflect wiring specificity after development. Together, our results suggest that while PV-INs differ in anatomy and in vivo activity, their continuous transcriptomic and homogenous biophysical landscapes are not predictive of these distinct identities. The relationship between gene expression and morphology to classify PV interneurons is unclear. Here, the authors show transcriptional continuity of morphologically distinct mouse hippocampal PV interneurons subtypes, combining single-cell RNA sequencing and electrophysiology."

(Fernandez et al., 2024)

Understanding OLM Interneurons: Characterization, Circuitry, and Significance in Memory and Navigation.

"Understanding the intricate mechanisms underlying memory formation and retention relies on unraveling how the hippocampus, a structure fundamental for memory acquisition, is organized. Within the complex hippocampal network, interneurons play a crucial role in orchestrating memory processes. Among these interneurons, Oriens-Lacunosum Moleculare (OLM) cells emerge as key regulators, governing the flow of information to CA1 pyramidal cells. In this review, we explore OLM interneurons in detail, describing their mechanisms and effects on memory processing, particularly in spatial and contextual memory tasks. Our aim is to provide a detailed understanding of how OLM interneurons contribute to the dynamic landscape of memory formation and retrieval."

Molecular Markers and Gene Expression

O-LM neurons are primarily characterized by their expression of somatostatin (SST) as the most consistent molecular marker, though they also express additional markers including neuropeptide Y (NPY), TRPV1, and the nicotinic acetylcholine receptor alpha2 subunit (Chrna2). Single-cell transcriptomics reveals these neurons form a relatively homogeneous population with some molecular heterogeneity and overlap in marker expression patterns. (15 sources)

The molecular identity of O-LM neurons has been established through extensive characterization of their gene expression patterns and protein markers. Somatostatin (SST) serves as the primary and most reliable molecular marker for identifying O-LM neurons across studies (Friend et al., 2019) (Oliva et al., 2000) (Winterer et al., 2019) (Hurtado-Zavala et al., 2017) (Nichol et al., 2018) (Bezaire et al., 2016) (Hooft et al., 2000). These somatostatin-positive neurons can be found throughout the stratum oriens, with particularly high densities near the alveus border and dense SST-positive neuropil observed in the stratum lacunosum-moleculare corresponding to their axonal projections (Lagartos-Donate et al., 2019) (Pinatel et al., 2023).

Beyond somatostatin, O-LM neurons express several additional molecular markers that provide more specific identification tools. The nicotinic acetylcholine receptor alpha2 subunit (Chrna2) has emerged as a particularly precise marker for O-LM interneurons, with expression restricted to the stratum oriens in both rats and mice and specific to this interneuron subtype (Thulin et al., 2025) (Leao et al., 2012) (Nichol et al., 2018). Single-cell transcriptomic analyses have revealed surprisingly consistent expression of neuropeptide Y (NPY) in O-LM neurons, along with co-expression of prepronociceptin (Pnoc) in transcriptomic clusters that define O-LM cells (Winterer et al., 2019) (Thulin et al., 2025).

O-LM neurons also express several functionally relevant receptors and enzymes that relate to their physiological properties. A significant proportion of morphologically-identified O-LM neurons express TRPV1 channels, with 81% of TRPV1-expressing neurons in stratum oriens also showing dense mGluR7 signal and co-expressing the inhibitory interneuron marker GAD65 (Hurtado-Zavala et al., 2017). These neurons express both endocannabinoid biosynthetic enzymes and type I metabotropic glutamate receptors necessary for endocannabinoid production, including diacylglycerol lipase alpha confirmed through immunohistochemical staining (Friend et al., 2019) (Oren et al., 2009).

The metabotropic glutamate receptor expression pattern is particularly distinctive, with O-LM neurons showing strong responses to mGluR activation and expressing both mGluR1 and mGluR5 (Hooft et al., 2000) (McBain et al., 1994). Some O-LM neurons also express mGluR1a along with somatostatin, though they are typically negative for other common interneuron markers including parvalbumin (PV), calbindin (CB), calretinin (CR), nitric oxide synthase (NOS), and various other markers (Katona et al., 2017).

Importantly, recent studies have revealed that molecular marker expression can overlap in morphologically-defined O-LM interneurons, with some neurons co-expressing PV, SST, or NPY (Hernandez-Frausto et al., 2023). Single-cell RT-PCR classification has identified O-LM neurons within broader categories that include SOM+, PV+, calretinin, and calbindin-expressing interneurons, though transcriptomic analyses suggest that O-

LM neurons constitute a relatively homogeneous population despite this molecular diversity (Friend et al., 2019) (Winterer et al., 2019). Interestingly, detailed transcriptomic analysis revealed infrequent expression of Htr3a in only approximately 10% of O-LM neurons, along with apparently specific expression of the 5-HT3b subunit-coding gene Htr3b in Htr3aCre-positive O-LM neurons (Winterer et al., 2019).

Evidence

(Friend et al., 2019)

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"SOM+ neurons have been previously shown to undergo LTP [32] in a manner that is dependent on type I mGluRs [32](Lapointe et al., 2004)(Vasseur et al., 2008)"

"RT-PCR results demonstrate that stratum oriens interneurons express mRNA for both endocannabinoid biosynthetic enzymes and the type I metabotropic glutamate receptors (mGluRs), necessary for endocannabinoid production. Immunohistochemical staining further confirmed the presence of diacylglycerol lipase alpha, an endocannabinoid-synthesizing enzyme, in oriens interneurons"

"Using single-cell RT-PCR, we classified four main groups: SOM+, PV+, calretinin, and calbindin, as previously reported (Freund et al., 1998)(Tamamaki et al., 2003)(Katsumaru et al., 2004)."

(Oliva et al., 2000)

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"EGFP was found to be expressed in a subpopulation of somatostatin-containing GABAergic interneurons in the hippocampus and neocortex"

"These newly identified interneuronal subtypes appeared to be closely related to O-LM cell, as they selectively innervate stratum lacunosum-moleculare. Whole-cell patch-clamp recordings revealed that these cells were fast-spiking and showed virtually no spike frequency accommodation. The microanatomical features of these cells suggest that they function primarily as "input-biasing" neurons, in that synaptic volleys in stratum radiatum would lead to their activation, which in turn would result in selective suppression of excitatory input from the entorhinal cortex onto CA1 pyramidal cells."

(Winterer et al., 2019)

Single-cell RNA-Seq characterization of anatomically identified OLM interneurons in different transgenic mouse lines

"oriens-lacunosum moleculare (OLM) interneurons. OLMs express somatostatin (Sst), generate feedback inhibition and play important roles in theta oscillations and fear encoding"

"cells that were included in the study were first reconstructed after DAB staining to identify the morphological characteristics of OLM cells, including the axonal projection to the lacunosum moleculare, the horizontally orientated soma and dendritic branching in the oriens"

"we found a surprisingly consistent expression of Npy in OLMs"

"our results revealed surprisingly infrequent expression of Htr3a in only ~10% of OLMs and an apparently specific expression of the 5-HT3b subunit-coding gene Htr3b in Htr3aCre-OLMs"

"OLMs constitute a highly homogenous transcriptomic population"

(Hurtado-Zavala et al., 2017)

TRPV1 regulates excitatory innervation of OLM neurons in the hippocampus

"These neurons have large somata with axons extending to the stratum lacunosum moleculare (Mu□ llet et al., 2014)(Cajal, 0) , and horizontal dendrites projecting through the stratum oriens as well as to further layers of the CA1 region (Freund et al., 1998) . Indeed,

we found that most TRPV1-expressing neurons in the stratum oriens also expressed somatostatin"

"TRPV1 is expressed in oriens-lacunosum-moleculare neurons"

"these markers co-localized in a sub-population of presumptive OLM cells in the stratum oriens"

"We found that 81% of TRPV1-expressing neurons in the stratum oriens are densely decorated by detectable mGluR7 signal"

"indicating that these neurons correspond to OLM neurons in the hippocampus. Most TRPV1-positive neurons were also positive for the inhibitory interneuron marker GAD65"

(Nichol et al., 2018)

Electrophysiological and Morphological Characterization of Chrna2 Cells in the Subiculum and CA1 of the Hippocampus: An Optogenetic Investigation

"The nicotinic acetylcholine receptor alpha2 subunit (Chrna2) is a specific marker for oriens lacunosum-moleculare (OLM) interneurons in the dorsal CA1 region of the hippocampus"

"These CA1 GABAergic, somatostatin (Som)-expressing interneurons are named for their distinctive morphology: their soma and dendritic trees are located in the stratum oriens and their axons extend directly out to arborize in the stratum lacunosum-moleculare (SLM; Cajal, 1911;(McBain et al., 1994)(Sik et al., 1995)(Maccaferri et al., 2000)(Losonczy et al., 2002)(Leão et al., 2012)"

"Chrna2 was also found to be expressed in Som-expressing cells of the subiculum (Leão et al., 2012)"

"We found that subiculum and CA1 dorsal Chrna2 cells similarly expressed the marker somatostatin and had comparable membrane and firing properties. The somas of Chrna2 cells in both regions were found in the deepest layer with axons projecting superficially. However, subiculum Chrna2 cells displayed more extensive projections with dendrites which occupied a significantly larger area than in CA1. The post-synaptic responses elicited by Chrna2 cells in pyramidal cells of both regions revealed comparable inhibitory responses elicited by GABAA receptors and, interestingly, GABAB receptor mediated components."

(Bezaire et al., 2016)

Interneuronal mechanisms of hippocampal theta oscillations in a full-scale model of the rodent CA1 circuit

"oriens-lacunosum moleculare (O-LM) cells (these SOM+ cells project to the distal dendrites in the stratum lacunosum-moleculare though their somata are located in the stratum oriens)"

(Hooft et al., 2000)

Differential Expression of Group I Metabotropic Glutamate Receptors in Functionally Distinct Hippocampal Interneurons

"Type I interneurons are located at the oriens-alveus border and have large, horizontally located cell bodies. Their dendritic trees run horizontally, and their axons run vertically toward stratum lacunosum-moleculare (Fig. 2 A). These interneurons contain somatostatin (Fig. 2 B). Therefore, these interneurons appear identical to the horizontal O-LM cells (Freund et al., 1998)(Katona et al., 1999)"

"Type I interneurons had large horizontally oriented cell somata located at the border of stratum oriens and the alveus, indicating that these cells were most likely identical with the previously described somatostatin-positive oriens-lacunosum moleculare (O-LM) cells (Freund et al., 1998). Reconstruction of type I interneurons revealed their horizontally oriented dendritic tree in stratum oriens and their axonal arborizations in stratum lacunosum-moleculare (n = 5) (Fig. 2 A), and in situ hybridization for somatostatin showed that four of four cells were indeed positive for somatostatin (Fig. 2 B)"

"Type I interneurons responded with a large inward current of −224 pA, were positive for somatostatin, and the majority expressed both mGluR1 and mGluR5"

"Activation of group I mGluRs under current-clamp conditions increased spike frequency and resulted in rhythmic firing activity in type I and II, but not in type III and IV, interneurons."

(Lagartos-Donate et al., 2019)

Pathological and non-pathological aging, SAMP8 and SAMR1. What do hippocampal neuronal populations tell us?

"CR+ neurons was found in stratum pyramidale"

"the distribution of the CR+ neurons was modified in the younger SAMP8 mice. Contrary to 6 months-old SAMR1 mice, the density was higher in the stratum oriens, pyramidal and lacunosum-moleculare of 6months-old SAMP8 mice"

"in 16 months-old SAMP8 the density of CR+ neurons of the stratum oriens and lacunosum-moleculare decreased significant values"

"SST+ cells were mainly found close to the alveus in the stratum-oriens of CA1 of both SAMR1 and SAMP8"

"A high density of SST+ neuropil was observed in the stratum lacunosum-moleculare as well."

(Pinatel et al., 2023)

A class-specific effect of dysmyelination on the excitability of hippocampal interneurons

"SST O-LM cells projecting to the distal apical dendrites in the stratum lacunosum-moleculare"

"We observed SST-positive axons crossing the stratum radiatum in a perpendicular orientation and ramifying at the level of the stratum lacunosum-moleculare, both in the 4.1B KO mice (Figure 2J) as in the wild-type (Figure 2I). These particular SST axons are likely O-LM axons."

(Thulin et al., 2025)

Differentiated Presynaptic Input to OLM 2 Cells Along the Hippocampal Dorsoventral Axis: Implications for Hippocampal Microcircuit Function

"Oriens lacunosum-moleculare (OLM) interneurons, identified by the Chrna2 gene"

"The Chrna2 gene expression is restricted to the stratum oriens in the hippocampus in both rats and mice (Ishii et al., 2005) and is specifically expressed in a subset of CA1 hippocampal interneurons, the oriens lacunosummoleculare (OLM) cells (Leão et al., 2012). Traditionally, OLM cells have been identified through their expression of somatostatin (Sst). However, in-depth single-cell transcriptomic cluster analysis has unveiled at least 11 distinct subpopulations of Sst-expressing interneurons (2017). Within these clusters, various classes of interneurons were identified, including back projecting, hippocampo-septal, oriens-bistratified, and OLM cells. Among these clusters, OLM cells were classified into a Sst and Prepronociceptin (Pnoc) co-expressing group (further divided into three subclusters)"

"OLM 2 cells receive differential innervation across the dorsal, intermediate, and ventral hippocampus"

"CA1 and CA3 inputs differ between hippocampal poles, suggesting that OLM 2 cells may have distinct feedback and feed-forward inhibitory roles in the hippocampal microcircuit. Intermediate OLM 2 cells uniquely receive substantial input from the subiculum and dorsal/medial raphe nuclei, as well as widespread CA2 inputs potentially linked to social memory. The medial septum and diagonal band of Broca provide cholinergic, GABAergic, and glutamatergic inputs across the axis, likely influencing disinhibition and oscillatory activity during various behavioral states. Excitatory input to intermediate-ventral OLM 2 cells partly arises from CA1 projection neurons targeting the mPFC."

(Leao et al., 2012)

OLM interneurons differentially modulate CA3 and entorhinal inputs to hippocampal CA1 neurons

"The vast diversity of GABAergic interneurons is believed to endow hippocampal microcircuits with the required flexibility for memory encoding and retrieval. However, dissection of the functional roles of defined interneuron types has been hampered by the lack of cell-specific tools. We identified a precise molecular marker for a population of hippocampal GABAergic interneurons known as oriens lacunosum-moleculare (OLM) cells. By combining transgenic mice and optogenetic tools, we found that OLM cells are important for gating the information flow in CA1, facilitating the transmission of intrahippocampal information (from CA3) while reducing the influence of extrahippocampal inputs (from the entorhinal cortex). Furthermore, we found that OLM cells were interconnected by gap junctions, received direct cholinergic inputs from subcortical afferents and accounted for the effect of nicotine on synaptic plasticity of the Schaffer collateral pathway. Our results suggest that acetylcholine acting through OLM cells can control the mnemonic processes executed by the hippocampus."

(Oren et al., 2009)

Role of Ionotropic Glutamate Receptors in Long-Term Potentiation in Rat Hippocampal CA1 Oriens-Lacunosum Moleculare Interneurons

"Some interneurons of the hippocampus exhibit NMDA receptor-independent long-term potentiation (LTP) that is induced by presynaptic glutamate release when the postsynaptic membrane potential is hyperpolarized. This "anti-Hebbian" form of LTP is prevented by postsynaptic depolarization or by blocking AMPA and kainate receptors. Although both AMPA and kainate receptors are expressed in hippocampal interneurons, their relative roles in anti-Hebbian LTP are not known. Because interneuron diversity potentially conceals simple rules underlying different forms of plasticity, we focus on glutamatergic synapses onto a subset of interneurons with dendrites in stratum oriens and a main ascending axon that projects to stratum lacunosum moleculare, the oriens-lacunosum moleculare (O-LM) cells. We show that anti-Hebbian LTP in O-LM interneurons has consistent induction and expression properties, and is prevented by selective inhibition of AMPA receptors. The majority of the ionotropic glutamatergic synaptic current in these cells is mediated by inwardly rectifying Ca2+-permeable AMPA receptors. Although GluR5-containing kainate receptors contribute to synaptic currents at high stimulus frequency, they are not required for LTP induction. Glutamatergic synapses on O-LM cells thus behave in a homogeneous manner and exhibit LTP dependent on Ca2+-permeable AMPA receptors."

(McBain et al., 1994)

Activation of metabotropic glutamate receptors differentially affects two classes of hippocampal interneurons and potentiates

excitatory synaptic transmission

"Based on responses to metabotropic glutamate receptor (mGluR) activation, we have characterized two distinct classes of interneuron in stratum (st.) oriens of the CA1 region of hippocampus. One type of interneuron was strongly excited by 1S,3R-aminocyclopentane dicarboxylic acid (ACPD), responding with a large inward current accompanied by increased baseline noise and prominent current oscillations. A second interneuron population responded with a modest inward current with no changes in baseline noise. These two classes of responses persisted in the presence of tetrodotoxin and antagonists of ionotropic glutamate and GABA receptors, suggesting that the inward currents result from mGluRs on the interneurons themselves. The two physiologically defined cell types correspond to two distinct morphological cell types in st. oriens/alveus, distinguished by very different patterns of local axonal connections. Large oscillatory inward current responses were recorded predominantly from an interneuron type whose axons heavily innervated st. lacunosum. The more modest inward current response was generally found in interneurons whose axons innervated the somata and proximal dendrites of CA1 pyramidal neurons. These differences in physiology and local circuitry imply that activation of mGluRs in st. oriens will cause very strong excitation of interneurons synapsing in st. lacunosum, and weaker excitation of interneurons innervating pyramidal cells at the soma and proximal dendrites. These data suggest that each interneuron population has a specific role in hippocampal function, and that mGluR activation will affect the local circuit differently for each interneuron type. Metabotropic GluR activation also markedly enhanced the amplitudes of the evoked and spontaneous EPSCs received by all interneurons in the region, independent of changes in the postsynaptic holding current and with no change in the kinetics of the EPSC. In contrast to the enhancement of evoked and spontaneous EPSCs, miniature EPSCs recorded in the presence of tetrodotoxin were not increased. These data suggest that ACPD acts at a presynaptic site to potentiate the EPSC. Taken together, these results highlight an important modulatory role for metabotropic receptors located at sites both pre- and postsynaptic to CA1 st. oriens interneurons."

(Katona et al., 2017)

Behavior-dependent activity patterns of GABAergic long-range projecting neurons in the rat hippocampus

"According to soma location, dendritic distribution and molecular composition the CA1-DG back-projection neuron ZsB49b is similar to most long-range projecting GABAergic neuron types and O-LM cells. However, the dense axonal output of ZsB49b to strata oriens, lacunosum-moleculare and the DG molecular layer makes it different from any previously documented GABAergic cell"

"We found strata oriens, lacunosum moleculare and moleculare of DG to be most heavily innervated by the neuron when quantified as total axonal length and total number of varicosities per layer (Fig. 1D), in contrast to strata radiatum and pyramidale, which had very few boutons"

"the cell body and the dendrites of the neuron were immunopositive for two markers SOM and mGluR1a (Fig. 1B,C). The neuron was immunonegative for PV, CB, CR, NOS, M2R, GABA-A receptor subunit alpha-3, neuropeptide Y, vasoactive intestinal polypeptide (VIP), and cannabinoid receptor type 1"

"the lack of a long-range projection axon and the dentate-innervating axonal collaterals clearly differentiate it from the former and latter cells, respectively."

(Hernandez-Frausto et al., 2023)

Local and long-range GABAergic circuits in hippocampal area CA1 and their link to Alzheimer's disease

"the expression of molecular markers PV, SST, or NPY can overlap in morphologically-defined oriens-lacunosum moleculare (OLM) interneurons"

Transcriptomic Classification and Heterogeneity

Single-cell transcriptomic analyses reveal that O-LM neurons form a surprisingly homogeneous population despite some molecular diversity, with distinct subpopulations identified through co-expression patterns and functional differences including PV-expressing subsets that contribute to gamma oscillations rather than theta rhythms. (5 sources)

Recent advances in single-cell transcriptomics have provided unprecedented insights into the molecular classification and heterogeneity of O-LM neurons, revealing both their fundamental unity and subtle diversity as a cell type. Despite traditional identification through somatostatin expression, in-depth single-cell transcriptomic cluster analysis has unveiled at least 11 distinct subpopulations of SST-expressing interneurons (Thulin et al., 2025). Within these broader SST-positive clusters, O-LM cells have been specifically classified into a somatostatin and prepronociceptin (Pnoc) co-expressing group that is further divided into three subclusters (Thulin et al., 2025).

Remarkably, morphologically-identified O-LM neurons display surprisingly consistent transcriptomic profiles, with studies revealing a surprisingly consistent expression of neuropeptide Y (NPY) across the population

(Winterer et al., 2019). This transcriptomic consistency extends to other molecular markers, though some heterogeneity exists in the expression of serotonin receptor subunits, with infrequent expression of Htr3a in only approximately 10% of O-LM neurons and apparently specific expression of the 5-HT3b subunit-coding gene Htr3b in Htr3aCre-positive O-LM neurons (Winterer et al., 2019).

Despite this overall molecular consistency, O-LM neurons exhibit functionally relevant heterogeneity that has important implications for network function. The O-LM cell class contains a functionally distinct subpopulation that co-expresses parvalbumin (PV), which represents a critical distinction for understanding their roles in network coordination (Zhang et al., 2025) (Fernandez et al., 2024). While SST-positive O-LM cells are primarily associated with theta rhythms, the PV-positive subset can fire in the gamma range, suggesting different roles in network coordination (Zhang et al., 2025) (Fernandez et al., 2024).

The transcriptomic analysis has also revealed spatial heterogeneity along the hippocampal axis, with O-LM neurons identified by Chrna2 gene expression receiving differential innervation across the dorsal, intermediate, and ventral hippocampus (Thulin et al., 2025) (Leao et al., 2012). This positional variation suggests that O-LM cells may have distinct feedback and feed-forward inhibitory roles in different parts of the hippocampal microcircuit, with intermediate O-LM cells uniquely receiving substantial input from the subiculum and dorsal/medial raphe nuclei, as well as widespread CA2 inputs potentially linked to social memory (Thulin et al., 2025).

Overall, transcriptomic studies demonstrate that O-LM neurons constitute a highly homogeneous transcriptomic population while maintaining functionally important subpopulations that contribute to the diversity of inhibitory control mechanisms within hippocampal circuits (Winterer et al., 2019).

Evidence

(Thulin et al., 2025)

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(Winterer et al., 2019)

Single-cell RNA-Seq characterization of anatomically identified OLM interneurons in different transgenic mouse lines

"oriens-lacunosum moleculare (OLM) interneurons. OLMs express somatostatin (Sst), generate feedback inhibition and play important roles in theta oscillations and fear encoding"

"cells that were included in the study were first reconstructed after DAB staining to identify the morphological characteristics of OLM cells, including the axonal projection to the lacunosum moleculare, the horizontally orientated soma and dendritic branching in the oriens"

"we found a surprisingly consistent expression of Npy in OLMs"

"our results revealed surprisingly infrequent expression of Htr3a in only ~10% of OLMs and an apparently specific expression of the 5-HT3b subunit-coding gene Htr3b in Htr3aCre-OLMs"

"OLMs constitute a highly homogenous transcriptomic population"

(Zhang et al., 2025)

Rethinking parvalbumin: From passive marker to active modulator of hippocampal circuits

"OLM cells have a distinct morphology and circuit position. Although canonically identified by the expression of somatostatin (SST), the OLM cell class is molecularly heterogeneous and contains a functionally distinct subpopulation that co-expresses PV (Fernández et al., 2024)(Que et al., 2021). This distinction is critical, as SST-positive OLM cells are primarily associated with theta rhythms, whereas the PV-positive subset can fire in the gamma range, suggesting a different role in network coordination. OLM cell bodies reside in the outermost layer of the hippocampus (stratum oriens), and they project a long axon up to the innermost layer (stratum lacunosum-moleculare, SLM), the most distal dendritic layer of pyramidal cells (Leão et al., 2012). This is precisely where the main excitatory input from the entorhinal cortex arrives via the perforant path. By selectively inhibiting these distal dendrites, OLM cells act as gatekeepers, controlling the flow of cortical information into the hippocampus and modulating dendritic integration and plasticity (Leão et al., 2012). They are critically involved in the generation of and phase-locking to network theta oscillations (4-12 Hz) (Amilhon et al., 2015)."

(Fernandez et al., 2024)

Understanding OLM Interneurons: Characterization, Circuitry, and Significance in Memory and Navigation.

"Understanding the intricate mechanisms underlying memory formation and retention relies on unraveling how the hippocampus, a structure fundamental for memory acquisition, is organized. Within the complex hippocampal network, interneurons play a crucial role in orchestrating memory processes. Among these interneurons, Oriens-Lacunosum Moleculare (OLM) cells emerge as key regulators, governing the flow of information to CA1 pyramidal cells. In this review, we explore OLM interneurons in detail, describing their mechanisms and effects on memory processing, particularly in spatial and contextual memory tasks. Our aim is to provide a detailed understanding of how OLM interneurons contribute to the dynamic landscape of memory formation and retrieval."

(Leao et al., 2012)

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